1. Basic file structure (e.g. workspaces) of ROS

The new build system for ROS is "catkin", while "rosbuild" is the old ROS build system which was replaced by catkin.

Packages in a catkin Workspace:

```
workspace_folder/          -- WORKSPACE
    src/                   -- SOURCE SPACE
        CMakeLists.txt     -- 'Toplevel' CMake file, provided by catkin
        package_1/
            CMakeLists.txt -- CMakeLists.txt file for package_1
            package.xml    -- Package manifest for package_1
        ...
        package_n/
            CMakeLists.txt -- CMakeLists.txt file for package_n
            package.xml    -- Package manifest for package_n
```

Create a workspace:

Code:

```
source /opt/ros/noetic/setup.bash
mkdir -p ~/catkin_ws/src
cd ~/catkin_ws/
```

catkin_make (Running it the first time in your workspace, it will create a CMakeLists.txt link in your 'src' folder.)

source devel/setup.bash (Before continuing, source your new setup.*sh file.)

Filesystem Tools:

i) Rospack: Rospack allows you to get information about packages.

Code: rospack find [package_name]

Would return: YOUR_INSTALL_PATH/share/roscpp

ii) Roscd: Roscd allows you to change directory (cd) directly to a package or a stack without mentioning the path.

Code: roscd <package-or-stack>

iii) Rosls: Rosls allows you to ls directly by name rather than it's absolute path.

Code: rosls <package-or-stack>

2. Operating procedure of ROS (Mention the key points, e.g. building the workspace, running roscore etc.)

i) Creating a catkin Package

cd ~/catkin_ws/src

catkin_create_pkg beginner_tutorials std_msgs rospy roscpp (Create a new package called 'beginner_tutorials' which depends on std_msgs, roscpp, and rospy. This will also create a beginner_tutorials folder.)

ii) Building a catkin workspace

cd ~/catkin_ws

catkin_make

. ~/catkin_ws/devel/setup.bash (To add the workspace to ROS environment)

iii) Starting ROS

roscore (Master + rosout + parameter server. Master is name service for ROS which helps nodes find each other. A node is an executable which uses ROS to communicate with other nodes. Rosout is ROS equivalent of stdout/stderr.)

3. Explain ROS nodes and how they are used to transmit data around ROS

A node really isn't much more than an executable file within a ROS package. Nodes can publish or subscribe to a Topic. Nodes can also provide or use a Service.

Rosnodes can be written in different programming languages –

- Rospy = python client library
- Roscpp = c++ client library

Using rosnode:

i) rosnode displays information about the ROS nodes that are currently running. The ‘rosnode list’ command lists these active nodes.

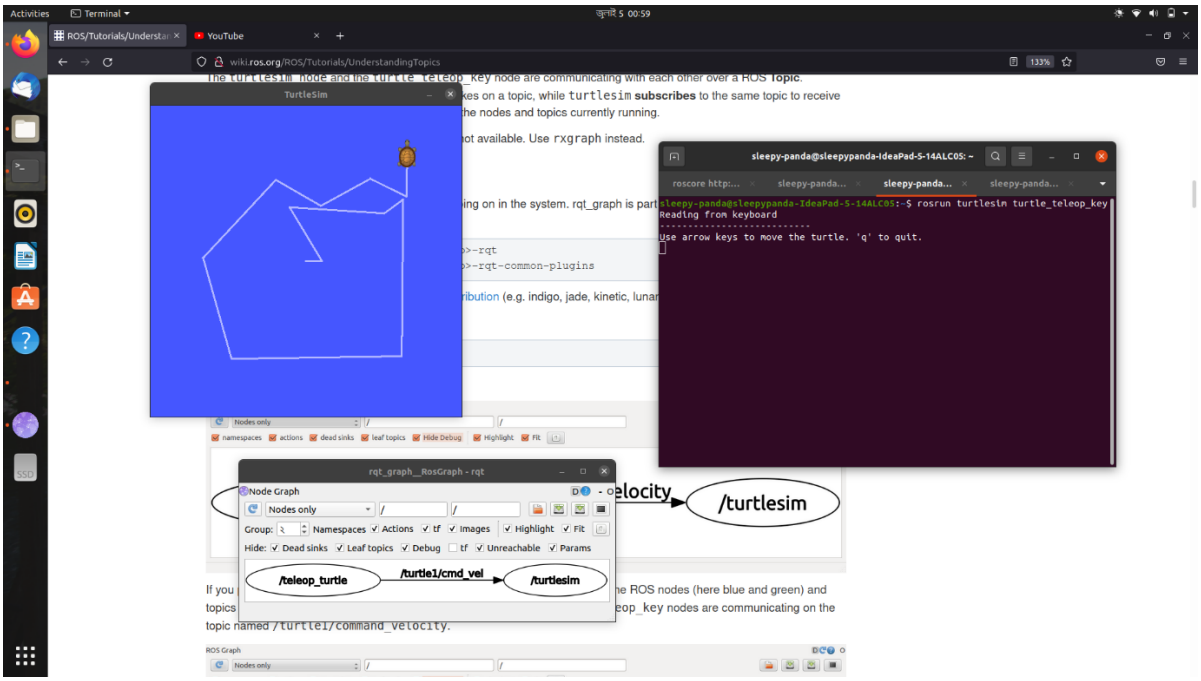
ii) The ‘rosnode info’ command returns information about a specific node.

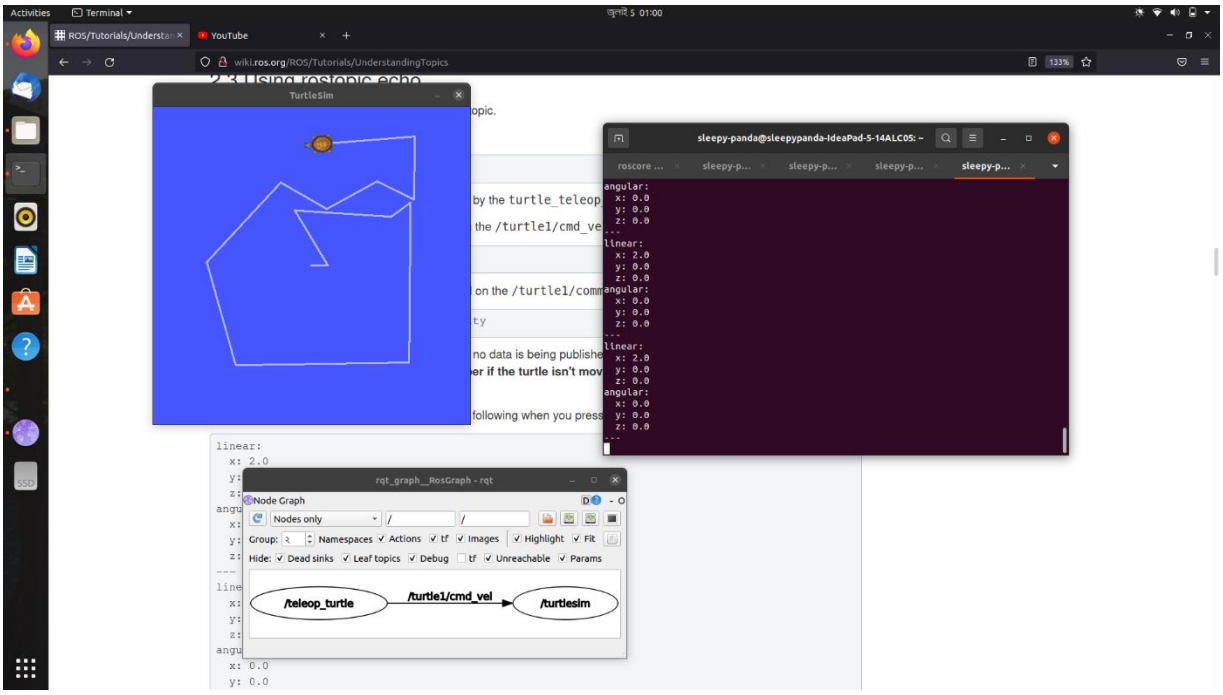
iii) rosrun allows you to use the package name to directly run a node within a package (without having to know the package path).

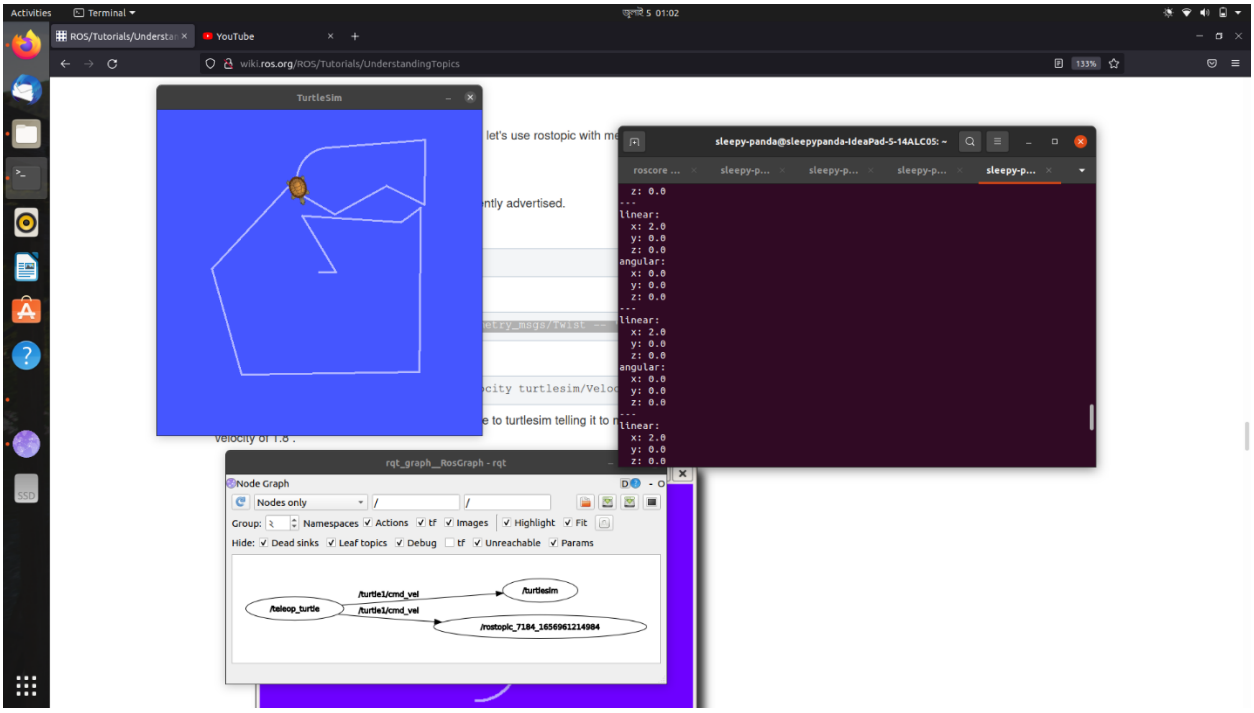
Code: rosrun [package_name] [node_name]

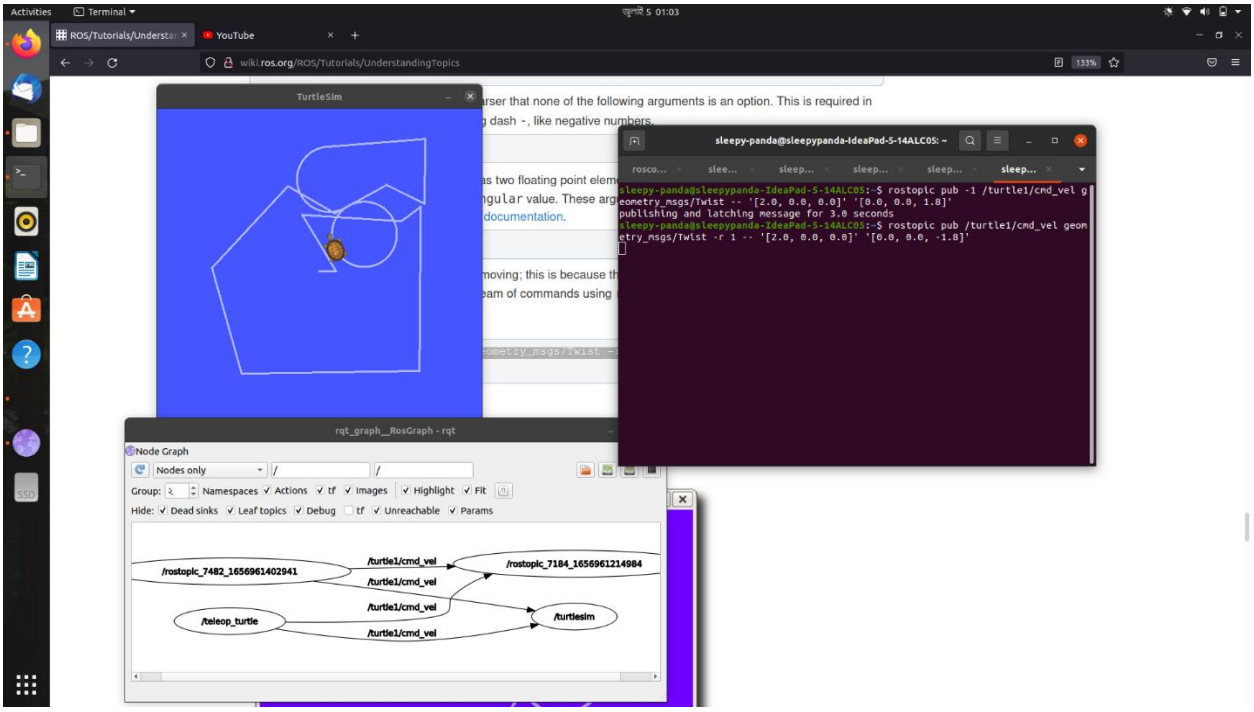
4. Some basic simulations in the online simulator

Using turtlebot simulation:


Activities
Terminal
ROS/Tutorials/Understar
YouTube
wiki.ros.org/ROS/Tutorials/UnderstandingTopics
TurtleSim
sleepy-panda@sleepypanda-IdeaPad-5-14ALC05: ~
sleepy-panda@sleepypanda-IdeaPad-5-14ALC05:~$ rosrun turtlesim turtle_teleop_key
Reading from keyboard
Use arrow keys to move the turtle. 'q' to quit.
rqt_graph__RosGraph - rqt
Node Graph
Nodes only
/teleop_turtle
/turtle1/cmd_vel
/turtlesim


Activities
Terminal
ROS/Tutorials/Understar
YouTube
wiki.ros.org/ROS/Tutorials/UnderstandingTopics
TurtleSim
sleepy-panda@sleepypanda-IdeaPad-5-14ALC05: ~
angular:
x: 0.0
y: 0.0
z: 0.0
---
linear:
x: 2.0
y: 0.0
z: 0.0
angular:
x: 0.0
y: 0.0
z: 0.0
rqt_graph__RosGraph - rqt
Node Graph
Nodes only
/teleop_turtle
/turtle1/cmd_vel
/turtlesim


Activities
Terminal
ROS/Tutorials/Understar
YouTube
wiki.ros.org/ROS/Tutorials/UnderstandingTopics
TurtleSim
sleepy-panda@sleepypanda-IdeaPad-5-14ALC05: ~
z: 0.0
---
linear:
x: 2.0
y: 0.0
z: 0.0
angular:
x: 0.0
y: 0.0
z: 0.0
rqt_graph__RosGraph - rqt
Node Graph
Nodes only
Group: Namespaces Actions tf Images Highlight Fit
Hide: Dead sinks Leaf topics Debug tf Unreachable Params
/turtle1/cmd_vel
/turtlesim
/teleop_turtle
/rostopic_7184_1656961214984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ROS/Tutorials/Understar
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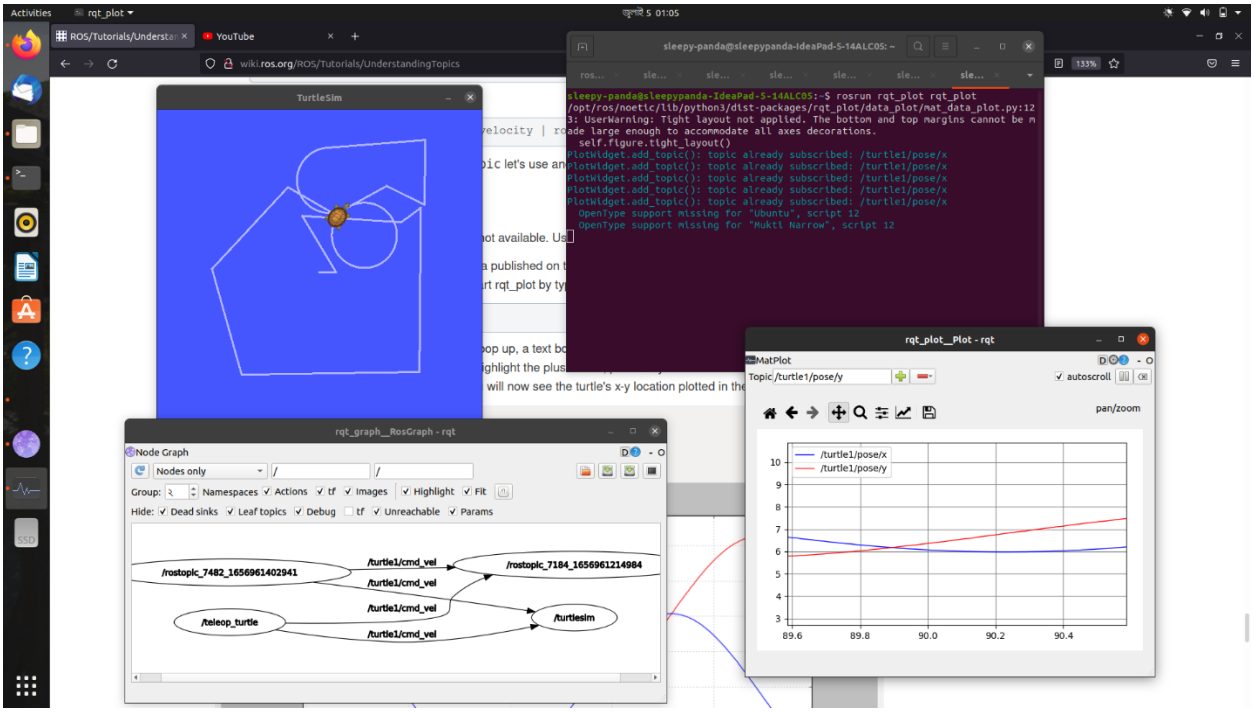
TurtleSim
sleepy-panda@sleepypanda-IdeaPad-5-14ALC05: ~
sleepy-panda@sleepypanda-IdeaPad-5-14ALC05:~$ rostopic pub -1 /turtle1/cmd_vel geometry_msgs/Twist -- '[2.0, 0.0, 0.0]' '[0.0, 0.0, 1.8]'
publishing and latching message for 3.0 seconds
sleepy-panda@sleepypanda-IdeaPad-5-14ALC05:~$ rostopic pub /turtle1/cmd_vel geometry_msgs/Twist -r 1 -- '[2.0, 0.0, 0.0]' '[0.0, 0.0, -1.8]'
rqt_graph__RosGraph - rqt
Node Graph
Nodes only
Group: Namespaces Actions tf Images Highlight Fit
Hide: Dead sinks Leaf topics Debug tf Unreachable Params
/rostopic_7482_1656961402941
/turtle1/cmd_vel
/rostopic_7184_1656961214984
/teleop_turtle
/turtlesim


Activities
rqt_plot
ROS/Tutorials/Understa
YouTube
wiki.ros.org/ROS/Tutorials/UnderstandingTopics
TurtleSim
sleepy-panda@sleepypanda-IdeaPad-5-14ALC05: ~
sleepy-panda@sleepypanda-IdeaPad-5-14ALC05:~$ rosrun rqt_plot rqt_plot
/opt/ros/noetic/lib/python3/dist-packages/rqt_plot/data_plot/mat_data_plot.py:12
3: UserWarning: Tight layout not applied. The bottom and top margins cannot be m
ade large enough to accommodate all axes decorations.
self.figure.tight_layout()
PlotWidget.add_topic(): topic already subscribed: /turtle1/pose/x
PlotWidget.add_topic(): topic already subscribed: /turtle1/pose/x
PlotWidget.add_topic(): topic already subscribed: /turtle1/pose/x
PlotWidget.add_topic(): topic already subscribed: /turtle1/pose/x
PlotWidget.add_topic(): topic already subscribed: /turtle1/pose/x
OpenType support missing for "Ubuntu", script 12
OpenType support missing for "Mukti Narrow", script 12
rqt_plot__Plot - rqt
MatPlot
Topic /turtle1/pose/y
autoscroll
pan/zoom
/turtle1/pose/x
/turtle1/pose/y
rqt_graph__RosGraph - rqt
Node Graph
Nodes only
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Namespaces
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tf
Images
Highlight
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Hide:
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Leaf topics
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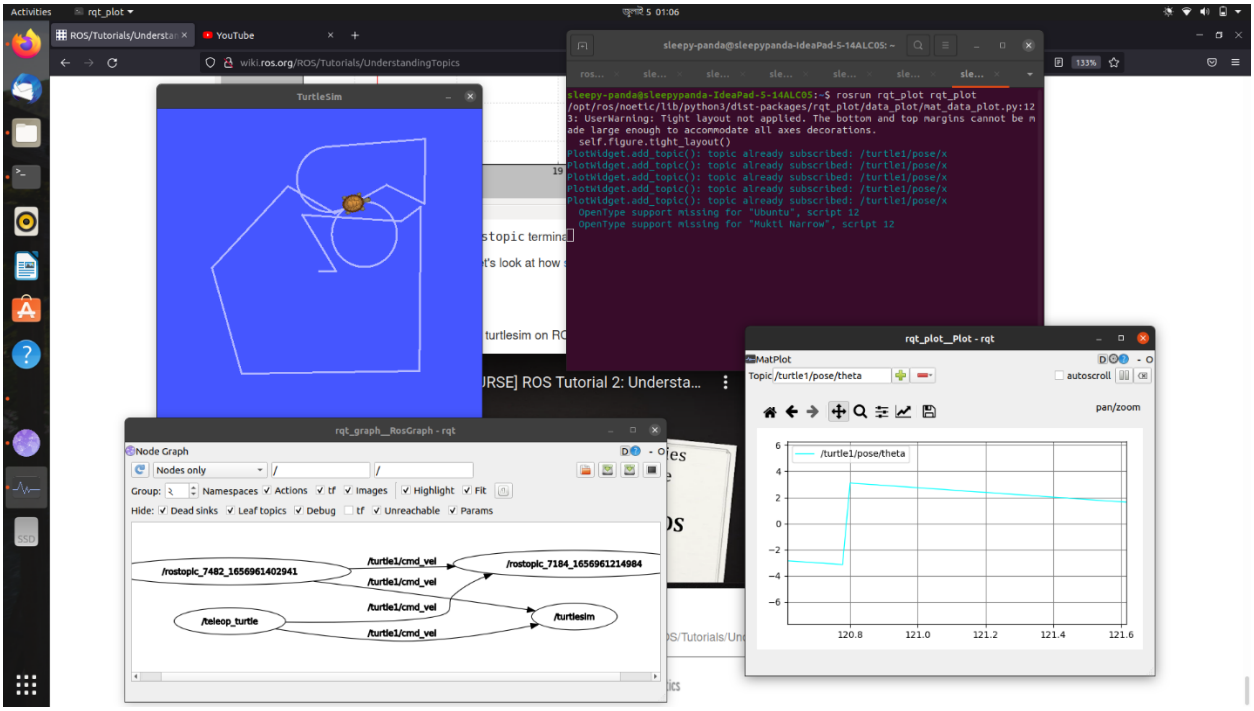
Unreachable
Params
/rostopic_7482_1656961402941
/rostopic_7184_1656961214984
/teleop_turtle
/turtlesim
/turtle1/cmd_vel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TurtleSim
sleepy-panda@sleepypanda-IdeaPad-5-14ALC05: ~
sleepy-panda@sleepypanda-IdeaPad-5-14ALC05:~$ rosrun rqt_plot rqt_plot
/opt/ros/noetic/lib/python3/dist-packages/rqt_plot/data_plot/mat_data_plot.py:12
3: UserWarning: Tight layout not applied. The bottom and top margins cannot be m
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self.figure.tight_layout()
PlotWidget.add_topic(): topic already subscribed: /turtle1/pose/x
PlotWidget.add_topic(): topic already subscribed: /turtle1/pose/x
PlotWidget.add_topic(): topic already subscribed: /turtle1/pose/x
PlotWidget.add_topic(): topic already subscribed: /turtle1/pose/x
PlotWidget.add_topic(): topic already subscribed: /turtle1/pose/x
OpenType support missing for "Ubuntu", script 12
OpenType support missing for "Mukti Narrow", script 12
rqt_plot__Plot - rqt
MatPlot
Topic /turtle1/pose/theta
autoscroll
pan/zoom
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/teleop_turtle
/turtlesim
/turtle1/cmd_vel

5. Shortcomings about ROS

i) I think a nominal GUI would have been nice for this operating system.

ii) Not having 100% dependency in Windows OS.

iii) I think it is pretty backdated to have unique dependencies in Arduino sketches (std:msgs and all those variable initialization things. It could have been made simpler by using C dependencies.)